

 Bennie-Kibet / Phase_4-project



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 README

Project: Twitter Sentiment Analysis

1. Business Understanding

1.1. Overview:

This project aims to build a Natural Language Processing (NLP) model capable of analyzing Twitter sentiments regarding Apple and Google products. The dataset consists of over 9,000 tweets that have been manually labeled as positive, negative, or neutral (neither). By analyzing customer opinions expressed on social media, organizations can gain insights into customer satisfaction, product perception, and emerging issues related to their products and services.

1.2. Problem Statement:

Organizations receive thousands of customer opinions daily through social media platforms such as Twitter. Manually analyzing these opinions is time-consuming and inefficient. Therefore, there is a need to develop an automated sentiment analysis model capable of accurately classifying tweets into positive, negative, or neutral sentiments to support data-driven decision-making and improve customer engagement strategies. To determine whether we can predict customer sentiments based on Tweets made on sold products.

1.3. Key Business Questions

- Can customer sentiment toward Apple and Google products be accurately predicted using tweet content?
- What proportion of tweets express positive, negative, or neutral sentiments?
- Which products receive the most positive and negative feedback?
- What words or phrases are most commonly associated with positive and negative sentiments?
- How can sentiment insights help organizations improve products and customer satisfaction?
- Can social media sentiment be used as an indicator of public perception and brand reputation?

1.4. Metrics:

From a business perspective:

- Sentiment Distribution – Percentage of positive, negative, and neutral tweets.
- Brand Perception Insights – Overall sentiment trends toward Apple and Google products.
- Customer Satisfaction Indicators – Proportion of positive versus negative opinions.
- Actionable Insights – Identification of recurring themes and concerns expressed by customers.

From a model perspective:

- Accuracy - To know how well our model is performing
- Confusion Matrix – Evaluates the classification performance across sentiment classes.

1.5 Objectives

Main Objective

To build a machine learning model capable of accurately classifying the sentiment of tweets related to Apple and Google products.

Specific Objectives

1. To explore and understand the characteristics of the Twitter sentiment dataset.
2. To preprocess tweet text through cleaning, tokenization, stopword removal, and text normalization.
3. To perform exploratory data analysis (EDA) to identify patterns and trends in customer sentiments.
4. To extract meaningful text features using techniques such as TF-IDF vectorization.
5. To train and evaluate machine learning models for sentiment classification.
6. To compare model performance using appropriate evaluation metrics.
7. To identify the most influential words and phrases contributing to positive and negative sentiments.
8. To provide business insights based on sentiment analysis results.

2. Data Understanding

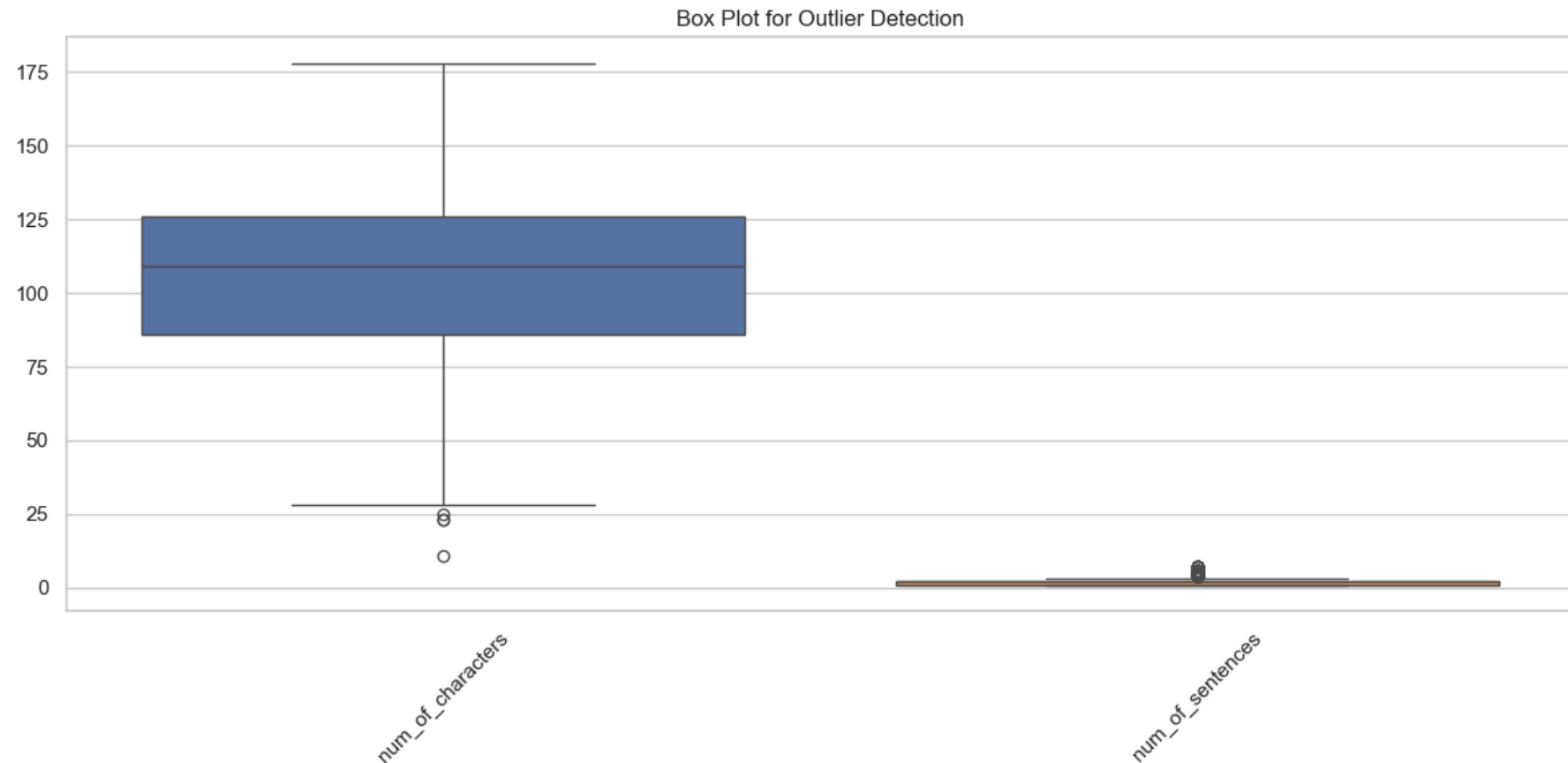
The data contains 9,093 data entries(rows) and is grouped under three features(columns).

The dataset features are:

1. tweet_text which contains the text entries of the users.
2. emotion_in_tweet_is_directed_at which contains the target product of the tweet_text, either Google or Apple.
3. is_there_an_emotion_directed_at_a_brand_or_product which contains the mood of the tweet_text on whether it is positive, negative or neither.

2.1 EXPLORATORY DATA ANALYSIS (EDA)

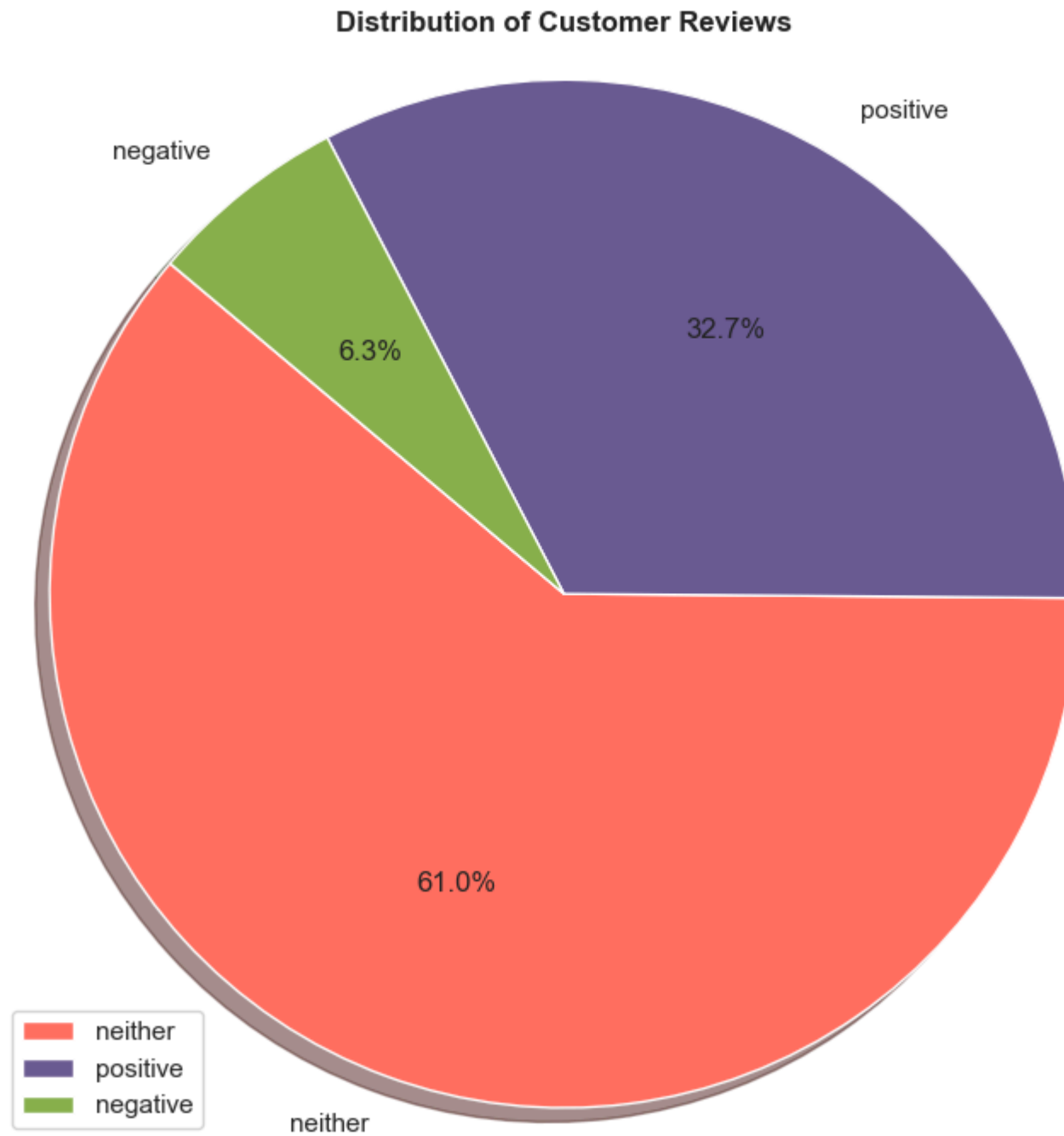
This stage involved checking for missing values and duplicates in the data and either dropping or filling them. This also involves checking for outliers which is done by the box plot below.



From the box plot above on the number of characters, majority of the tweets range from slightly above 25 characters to slightly above 175 characters with majority of the data clustered from slightly above 125 characters and between 75 and 100 characters. The box plot on the number of sentences, the number of sentences in the tweets are clustered between 1 and 5 with the majority of the number of sentences being 2 sentences.

2.2 Univariate Analysis

Comparing each variable at a time.

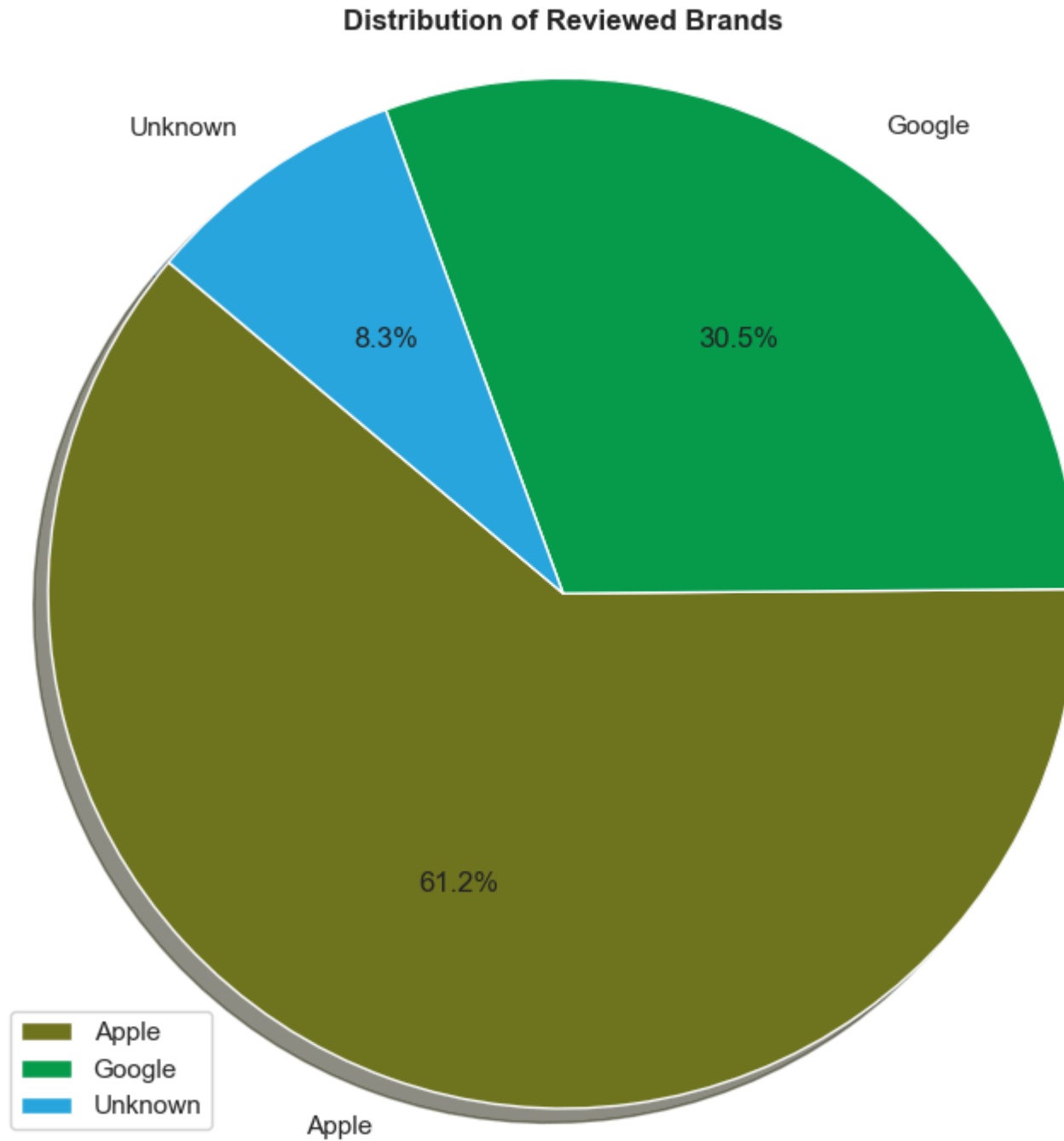


Key Insights

- Most customers gave **neutral reviews** (61%), indicating that many reviews are neither strongly positive nor negative.
- **Positive reviews** make up nearly one-third of the total, suggesting a generally favorable customer sentiment among those expressing an opinion.
- **Negative reviews** are relatively rare (6.3%), indicating low levels of dissatisfaction.

Overall Interpretation

The review distribution suggests that customer sentiment is largely **neutral**, with **positive feedback significantly outweighing negative feedback**. This could indicate that while many customers have moderate or mixed experiences, the overall perception is more favorable than unfavorable.



Key observations

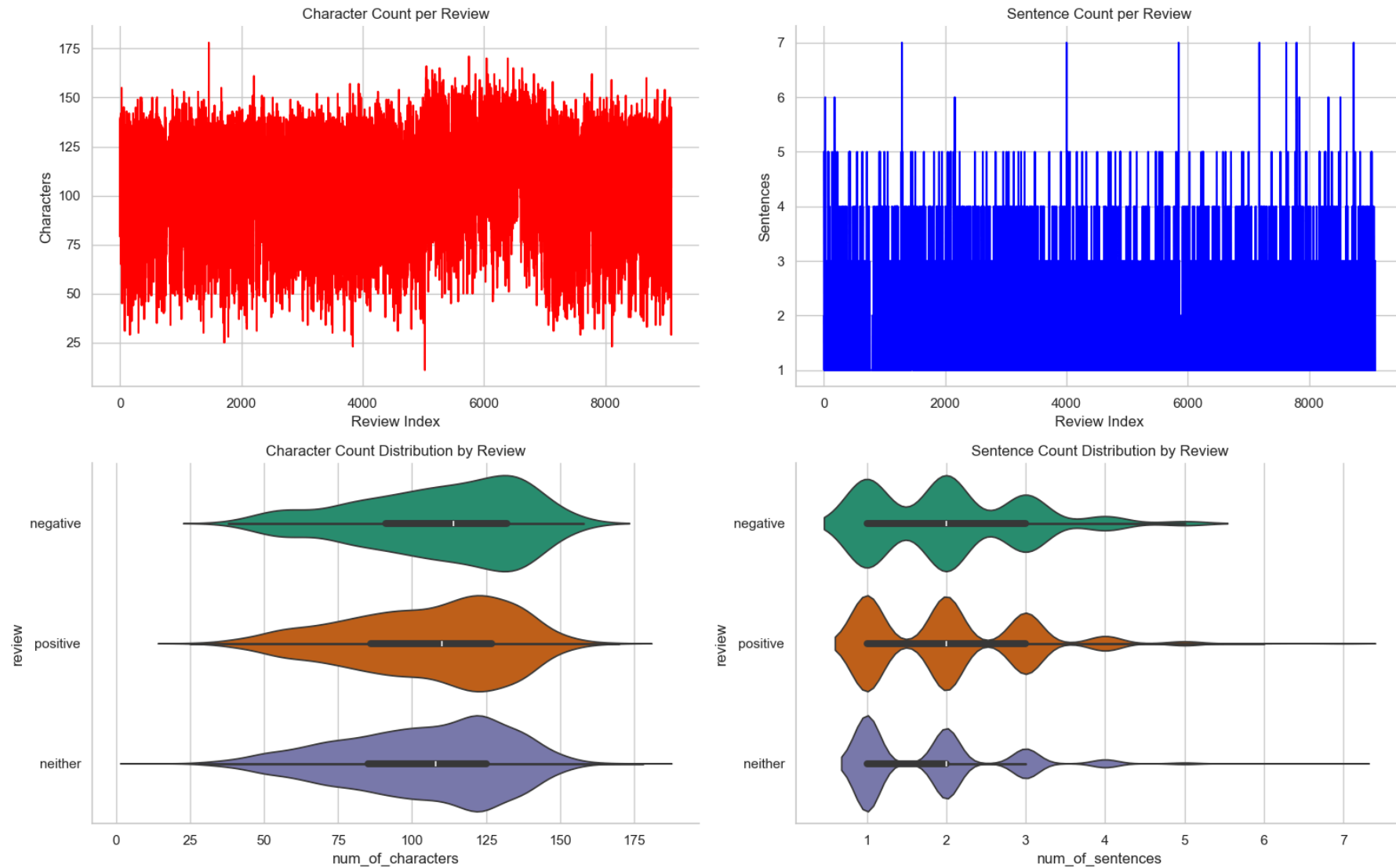
- Apple accounts for **more than three-fifths** of the total reviews.
- Apple and Google together make up **91.7%** of all reviewed brands, indicating that the dataset is heavily concentrated on these two brands.
- The **Unknown** category represents a relatively small portion of the reviews.

Overall interpretation

The reviewed brands are strongly skewed toward Apple, with Google being the second most frequently reviewed brand. Very few reviews could not be associated with a known brand. This suggests that the dataset is primarily focused on Apple and Google products or services.

2.3 Bivariate Analysis

In this part we evaluate relationship between the feature columns and the target column (review)



The figure presents a **multivariate exploratory analysis of text-length characteristics** across three sentiment classes (**positive, negative, and neither**). It consists of two line plots and two violin plots.

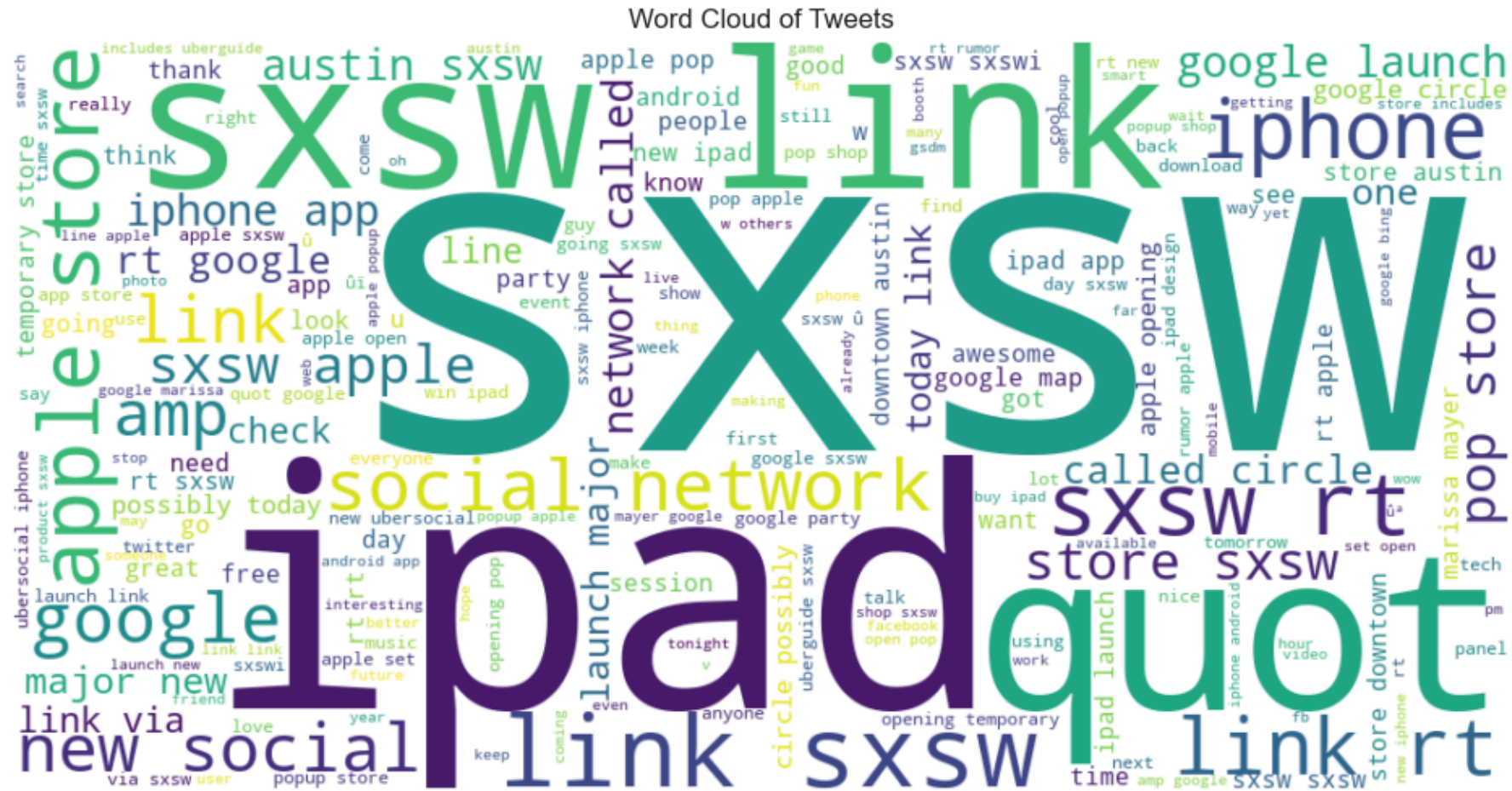
In summary:

- From the top left plot, the dataset contains moderately sized texts with no extreme variation, which is beneficial for NLP modeling since document lengths are fairly uniform.
- From the top right plot, the corpus is dominated by short reviews, which is typical for social media posts, comments, or brief product reviews.
- From the bottom left plot, the violin plots compare character count distributions across sentiment categories. The three sentiment classes exhibit very similar character-count distributions, this suggests that review length alone may not be a strong predictor of sentiment.
- From the bottom right plot, the violin plots illustrate the number of sentences across sentiment categories. Sentence count distributions overlap substantially across all sentiment categories. Therefore, sentence count is unlikely to be a highly discriminative feature for sentiment classification.

3. Data Preprocessing

This stage involved:

1. removing punctuations, numbers, and other symbols in the tweet_text using the function `re()`
2. lowering the case of the tweet_text using the function `lower()`
3. converting the tweet_text into tokens using `wordtokenize`
4. removing stopwords
5. lemmatizing the text
6. stemming the text



This combines **all tweets**, regardless of sentiment.

Key observations:

- The most frequent terms overall are: **sxsw**, **ipad**, **link**, **apple**, **store**, **iphone**, **google**, **social**, **network** and **launch**.
- The cloud reflects the main themes of the conversation:
 - Mobile devices (iPad, iPhone, Android)

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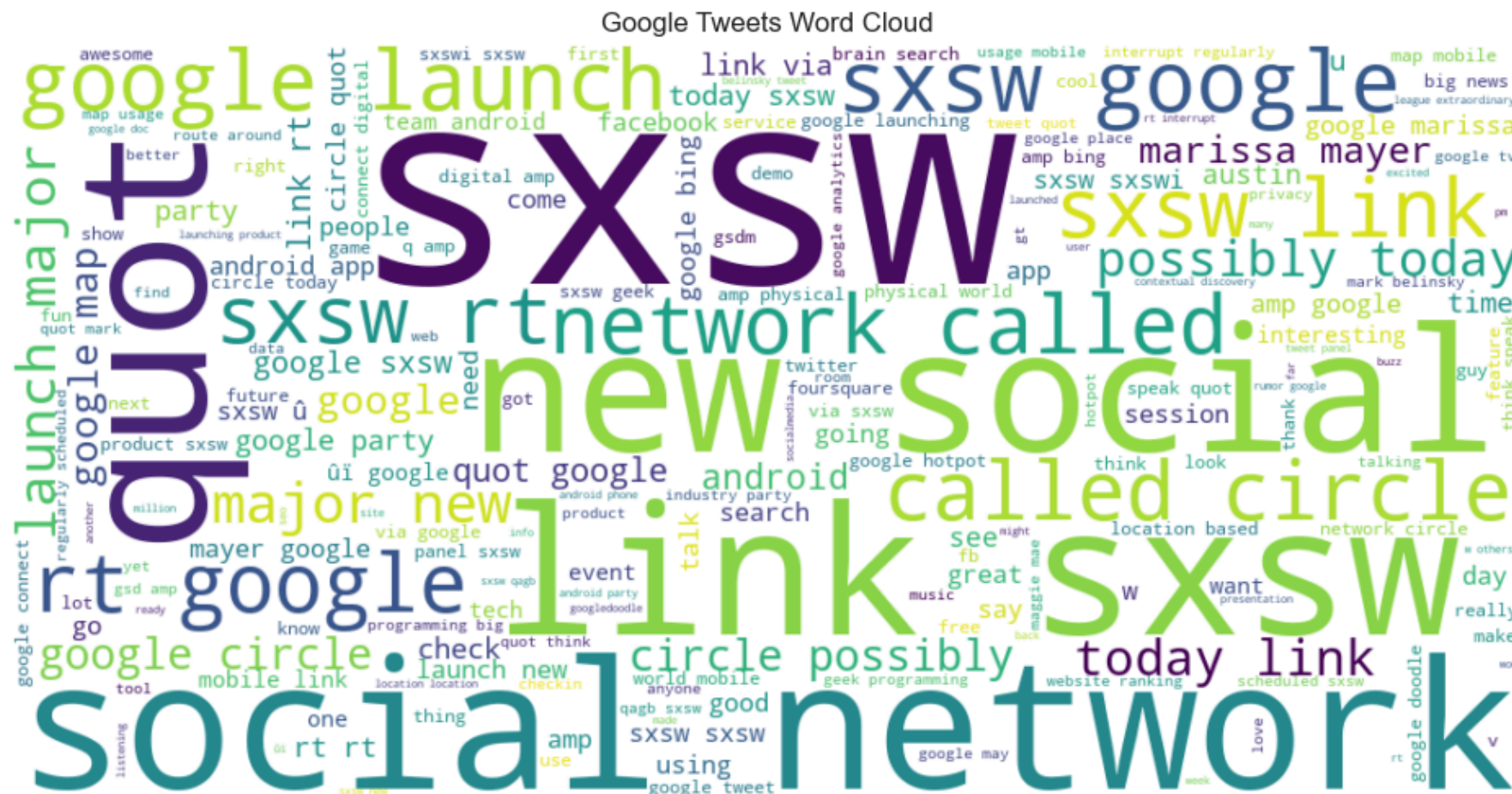
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- The most dominant words are **ipad**, **sxsw**, **apple**, **store**, **iphone**, **app**, and **link**.

- The prominence of **ipad** and **iphone** indicates that discussions are largely centered on Apple's flagship mobile devices.
- Frequent appearances of **store**, **opening**, **downtown**, and **austin** suggest significant discussion around Apple Store-related events.
- The term **sxsw** (South by Southwest) is highly dominant, indicating that many tweets were generated during or about the SXSW technology event.
- Words such as **launch**, **party**, **free**, and **pop** imply conversations related to product launches, promotions, and event activities.

In Summary:

The Apple tweets focus heavily on Apple products, especially the iPad and iPhone, with much of the conversation linked to promotional activities and events occurring during SXSW. The overall vocabulary reflects customer engagement with product showcases, store openings, and app-related discussions.



Key Observations

- The most prominent words are **"social"**, **"network"**, **"new"**, **"called"**, **"circle"**, **"google"**, **"launch"**, and **"sxsx"**.
- The phrase components **"social network called circle"** dominate the cloud, indicating extensive discussion about Google's social networking initiative, Google+ (originally rumored as Google Circle).
- Words such as **"launch"**, **"today"**, **"possibly"**, and **"called"** suggest speculation and anticipation surrounding a product announcement.
- Terms including **"android"**, **"search"**, and **"google"** indicate continued discussion of Google's core products and services.

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Key observations:

- Again, **sxsw** is the largest word.
- Strongly represented terms include; **google**, **quot** (likely an artifact from tweet text processing), **link**, **ipad**, **iphone**, **apple**, **store**, **great**, **awesome**, **free** and **good**.
- Positive tweets contain more explicitly favorable words such as: **awesome**, **great**, **good** and **nice**.

Interpretation:

Positive discussions revolve around technology launches, apps, Apple and Google products, conference activities, and excitement around SXSW-related announcements.



This cloud highlights the most frequent words in tweets classified as **negative**.

Key observations:

- **sxsw** is the dominant term, indicating the event is the main discussion topic.
- Other prominent words include; **ipad**, **apple**, **google**, **link**, **android**, **store**, **iphone**, **need** and **app**.
- Negative tweets appear to focus on:
 - Technical issues ("link", "app", "store")

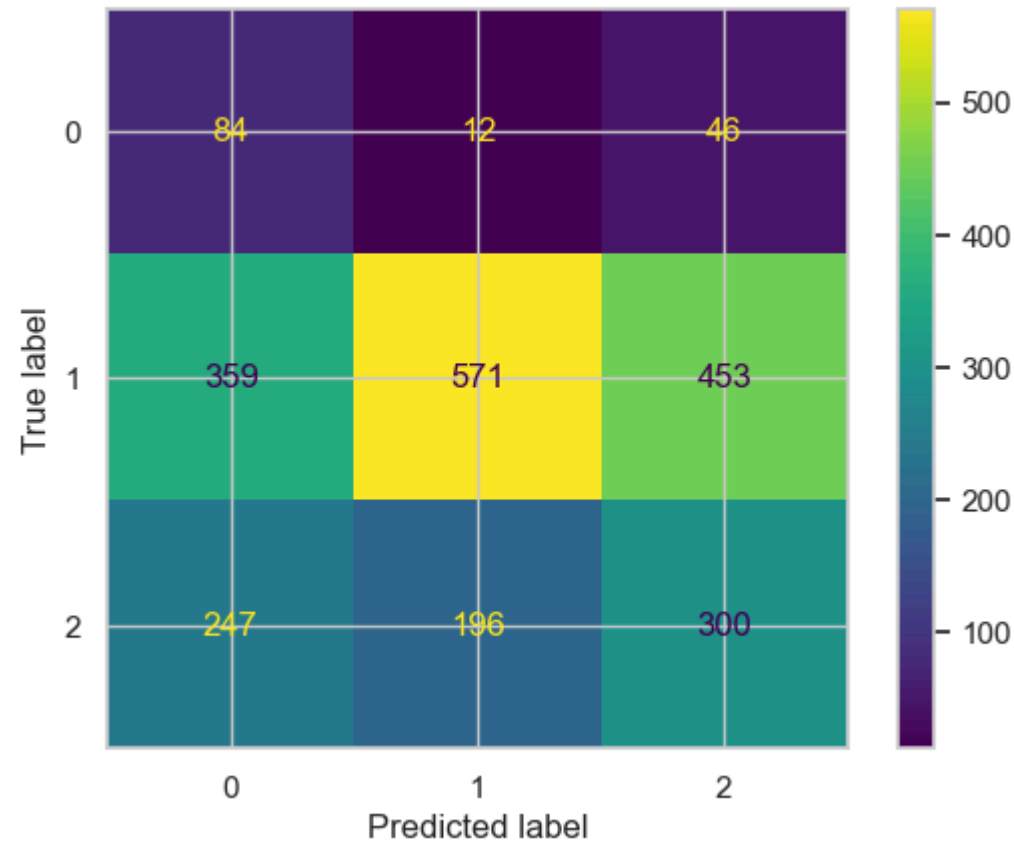
- Product discussions ("ipad", "iphone", "android")
- Frustrations or unmet expectations ("need", "fail", "problem"-related terms appearing in smaller sizes)

Interpretation:

Negative sentiment seems tied more to user experiences with apps, devices, links, or conference-related technology rather than strong criticism of a particular company.

4. Modeling

4.1. Baseline Model: Random Forest Model



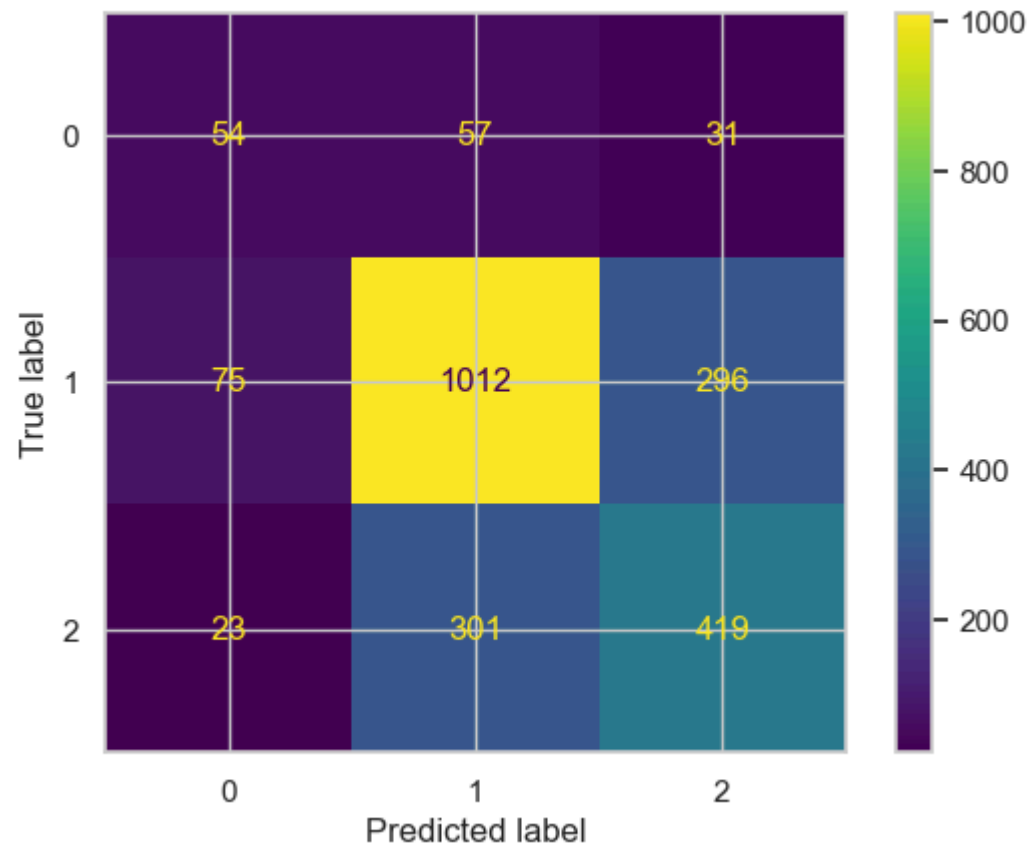
The model had an accuracy of 42% (0.4210758377425044). Performing poorly showing that it predicts correctly 42% of the time.

The values are:

- True 0: 84 / 12 / 46
- True 1: 359 / 571 / 453
- True 2: 247 / 196 / 300

Class 1 dominates the dataset by far (it has the most total samples), and while 571 of those are correctly predicted (the bright yellow cell), a large number leak into both class 0 (359) and class 2 (453). Class 0 has only 142 total samples and is correctly identified just 84 times, with notable confusion toward class 2. Class 2 (743 samples) is correctly classified only 300 times, with substantial spillover into both other classes (247 to class 0, 196 to class 1). Overall, the model shows weak separability, especially heavy bleed from class 1 into the other two classes.

4.1.2. Model 2: Optimized Random Forest Model



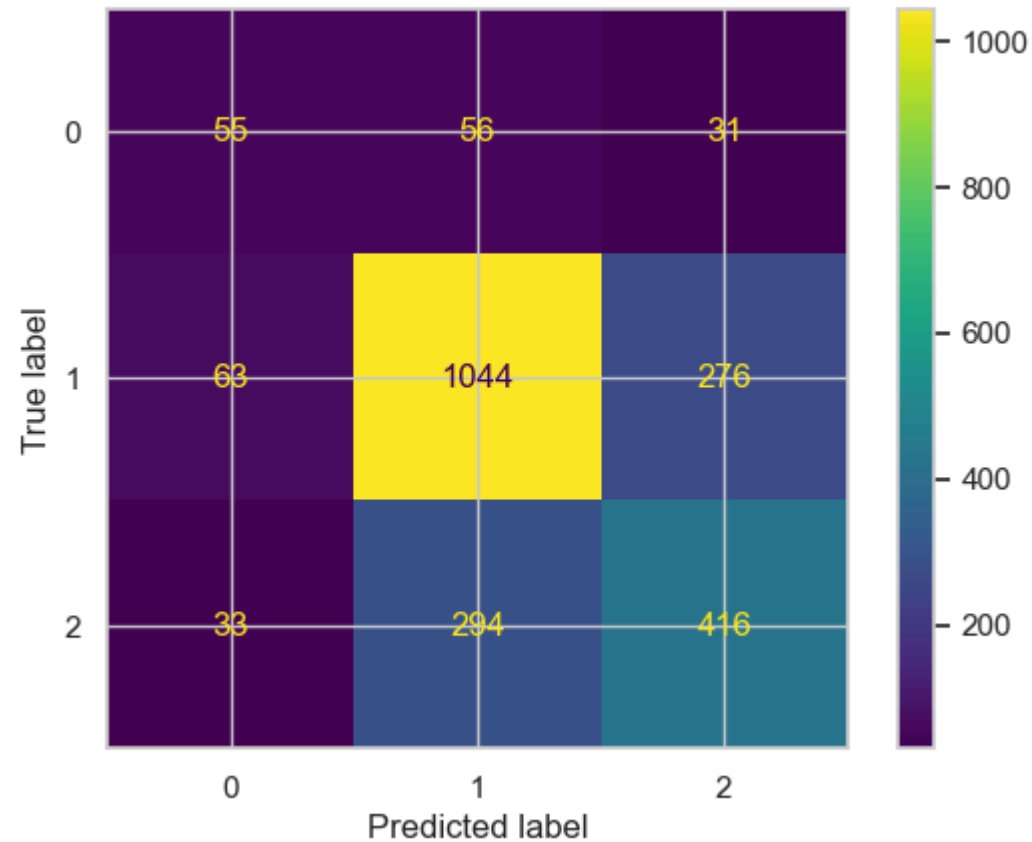
The tuned model performs better than the untuned model with an accuracy of 65% (0.6547619047619048). Showing it can predict correctly 65% of the time.

The values are:

- True 0: 54 / 57 / 31
- True 1: 75 / 1012 / 296
- True 2: 23 / 301 / 419

This matrix shows a much sharper diagonal for class 1, jumping from 571 to 1012 correct predictions, with confusion into class 0 nearly eliminated (359 → 75). Class 2's correct predictions also improved (300 → 419), and its confusion with class 0 dropped sharply (247 → 23), though confusion with class 1 remains similar (196 → 301). However, class 0 performance got worse: correct predictions dropped from 84 to 54, with more of its samples now misclassified as class 1 (12 → 57) while misclassification as class 2 stayed similar (46 → 31).

4.2. Model 3: Linear Support Vector Machine(SVM) Model



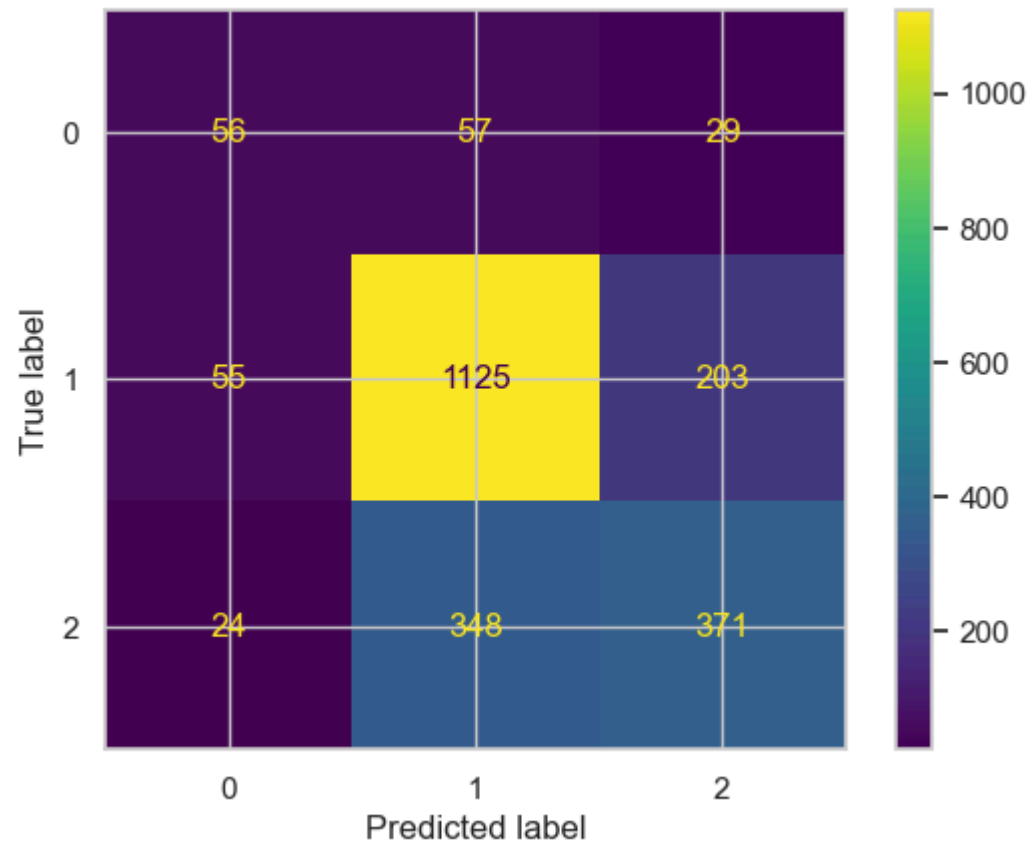
The model has an accuracy of 67% (0.667989417989418). Showing it can predict correctly 67% of the time on unseen data.

The values are:

- True 0: 55 / 56 / 31
- True 1: 63 / 1044 / 276
- True 2: 33 / 294 / 416

This matrix looks very similar in pattern to the previous "opt" model: class 1 (the majority class) is predicted correctly the vast majority of the time (1044/1383), with a notable chunk of class 1 samples misclassified as class 2 (276). Class 0 remains the weakest, with only 55 of 142 correctly identified and roughly equal confusion with classes 1 and 2 (56 and 31). Class 2 is correctly classified 416/743 times, with substantial confusion toward class 1 (294).

4.2.2. Model 4: Tuned Support Vector Machine(SVM) Model



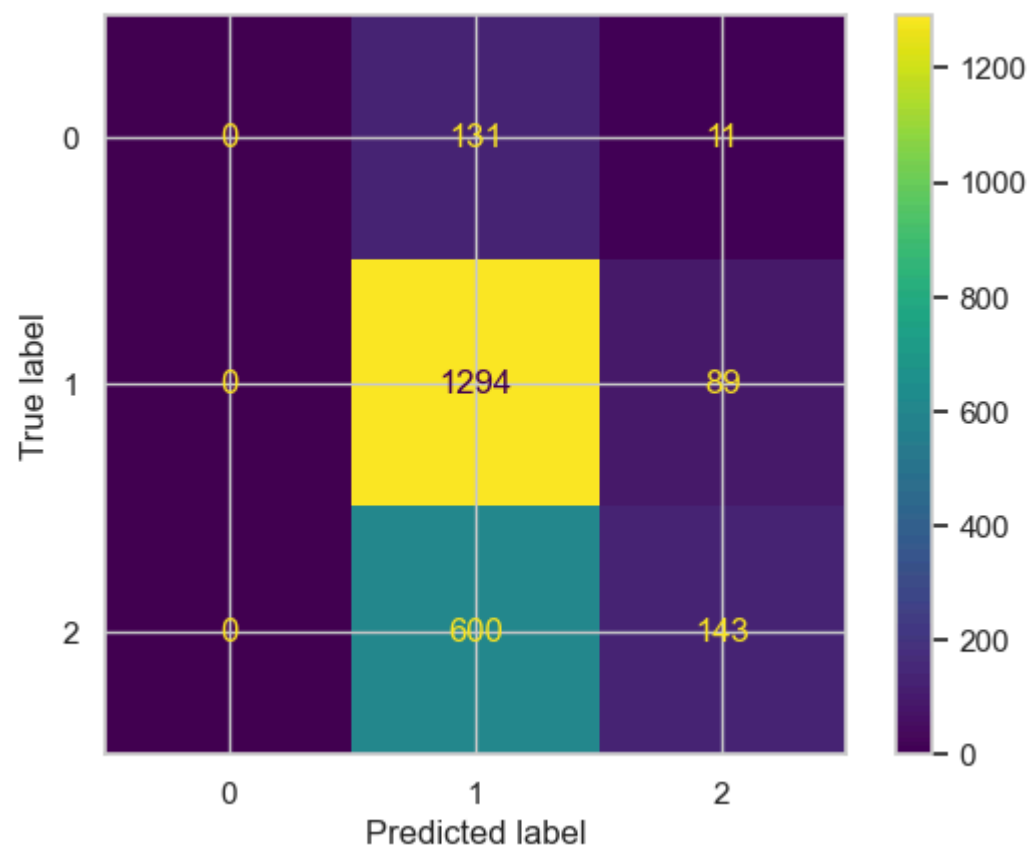
The model has an accuracy of 68% (0.6843033509700176). Showing it can predict correctly 68% of the time on unseen data.

The values are:

- True 0: 56 / 57 / 29
- True 1: 55 / 1125 / 203
- True 2: 24 / 348 / 371

The overall pattern is similar, but tuning shifted performance slightly: class 1's correct predictions increased from 1044 to 1125, and its confusion with class 2 dropped (276 → 203). However, class 2's own accuracy fell (416 → 371), with more of its samples now being misclassified as class 1 (294 → 348). Class 0 stayed essentially unchanged (55 → 56 correct).

4.3. Model 5: MultinomialNB Model



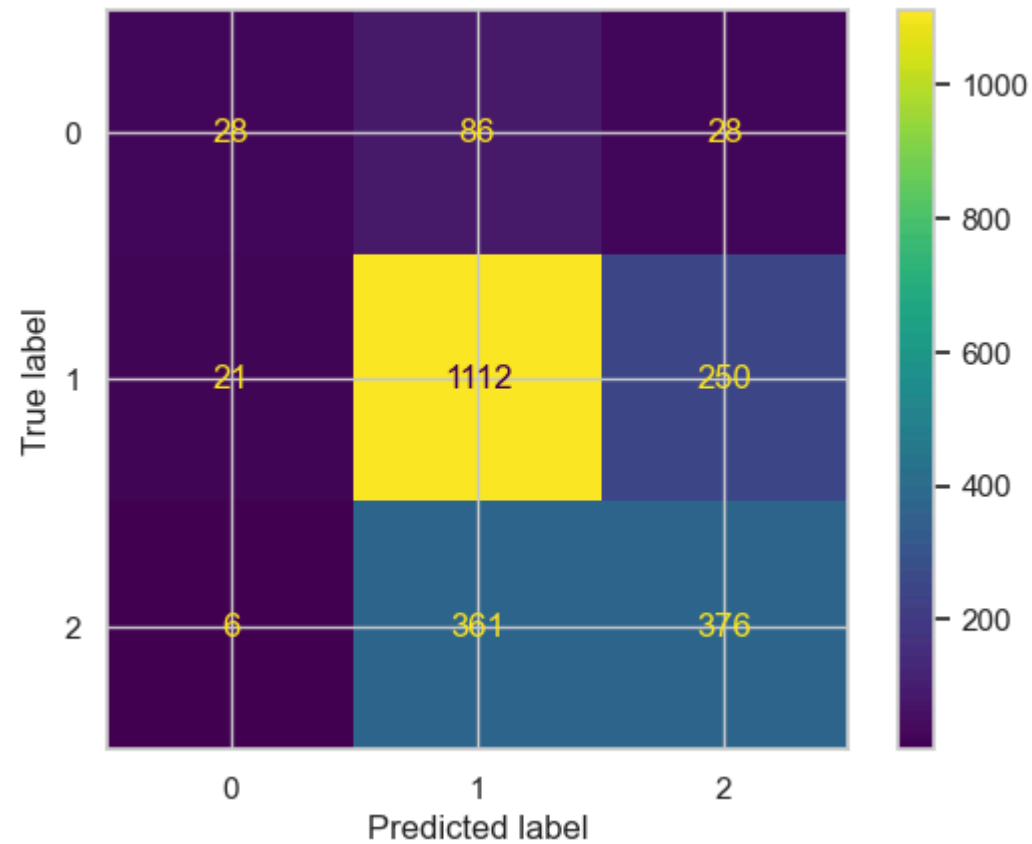
The model has an accuracy of 63% (0.6335978835978836). Showing it can predict correctly 63% of the time on unseen data.

The values are:

- True 0: 0 / 131 / 11
- True 1: 0 / 1294 / 89
- True 2: 0 / 600 / 143

As noted previously, the first column is entirely zero — this model never predicts class 0 at all, so every true class-0 sample is misclassified (mostly into class 1). Class 1 has very high recall (1294/1383), while class 2 is mostly absorbed into class 1 (600 of 743).

4.3.2. Model 6: Tuned MultiNomialNB Model



The model has an accuracy of 67% (0.6684303350970018). Showing it can predict correctly 67% of the time on unseen data.

The values are:

- True 0: 28 / 86 / 28
- True 1: 21 / 1112 / 250
- True 2: 6 / 361 / 376

Tuning has clearly changed the model's behavior: class 0 is now predicted at all (the first column is no longer empty), and 28 of the 142 true class-0 samples are now correctly identified. However, this comes with trade-offs elsewhere: class 1's correct predictions dropped from 1294 to 1112, with more spillover into class 2 (89 -> 250). Class 2's correct predictions increased from 143 to 376, with reduced (but still substantial) confusion toward class 1 (600 -> 361).

4.4. Model 7: KNearestNeighbours Model

